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Design of a Programmable Gain Amplifier
Using a Fully Differential OTA
with 5T OTA-Based CMFB and
DC Offset Cancellation

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Instructor: Prof. Abhishek Srivastava

Student Name: Siddarth Gottumukkula

Roll Number: 2023102040

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1 Introduction and Motivation

The design of precision analog front-ends in modern mixed-signal systems demands careful attention to gain accuracy, bandwidth programmability, dynamic range, and power efficiency. A *Programmable Gain Amplifier* (PGA) is a cornerstone block in receivers, sensor-interface circuits, and software-defined radio basebands, where the signal amplitude may vary over several decades and a fixed-gain amplifier would either saturate or fail to resolve weak signals. The PGA designed in this project targets a closed-loop gain range of 10 dB to 50 dB with a bandwidth of at least 750 kHz at maximum gain.

The core amplifying element is a *fully differential operational transconductance amplifier* (OTA). A differential topology is preferred over a single-ended one for three fundamental reasons: (i) the inherent rejection of common-mode disturbances, including supply noise, by a factor equal to the CMRR; (ii) a doubling of the output voltage swing for a given supply voltage, thereby improving dynamic range; and (iii) suppression of even-order harmonic distortion, which is critical for high-linearity applications. The target supply of 1.8 V (SCL 180 nm CMOS) leaves limited headroom, making the differential architecture essential for maximising usable signal swing.

A fully differential OTA, however, introduces the problem of *common-mode instability*. Unlike a single-ended amplifier, there is no inherent mechanism to define the output common-mode voltage; any mismatch or device variation will drift the outputs toward one rail. An active *Common-Mode Feedback* (CMFB) loop is therefore indispensable. This project implements a continuous-time CMFB using a second 5-transistor (5T) OTA, which senses the average of the differential outputs, compares it to a reference, and injects a correction current into the main amplifier's bias network. The choice of a 5T OTA for the CMFB block is motivated by its simplicity, low power consumption, and adequate loop bandwidth for stabilising the common-mode within the differential signal band.

At high closed-loop gains (up to 50 dB), even a modest input-referred DC offset of a few millivolts maps to hundreds of millivolts or more at the output, causing saturation and rendering the amplifier useless. A *DC Offset Cancellation* (DCOC) loop - implemented as a low-frequency integrator/feedback path - suppresses this offset to below 10 mV at the output, allowing the full dynamic range to be exploited. The gain programmability is achieved through a switched resistor-ladder feedback network.

This report documents the complete design flow: theoretical foundations and hand calculations (sections 3 and 4) and simulation methodology and results (section 5), followed by the conclusion (section 6). All simulations were performed in Cadence Virtuoso using the SCL 180 nm CMOS process design kit (PDK).

2 System Architecture

2.1 Top-Level Block Diagram

The proposed PGA system consists of four distinct functional blocks: the fully differential core OTA, the 5T-OTA-based continuous-time CMFB, the programmable switched-resistor feedback network, and the DC offset cancellation integrator loop. Fig 1 shows the top-level block diagram.

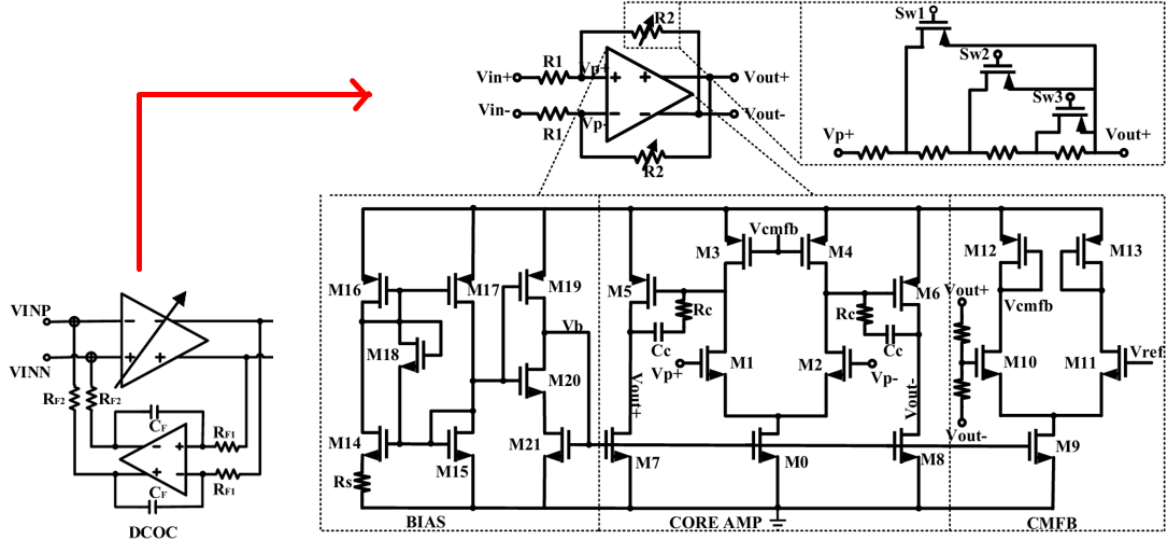


Figure 1: Top-level block diagram of the entire circuit.

2.2 Fully Differential Core OTA

The primary amplifying block is a differential-pair OTA with NMOS input transistors and an NMOS current-mirror active load. The architecture considered for the core amplifier consists of 2 stages - a 5T OTA followed by a CS amplifier. **5-Transistor (5T) OTA** is a simple differential OTA topology, consisting of a differential input pair, a current-mirror load, and a tail current source. Its single-pole dominant nature and low transistor count make it power-efficient. To meet the gain specification, a CS amplifier is used and it is Miller compensated for stability.

2.3 CMFB

The CMFB block must sense the output common-mode voltage $V_{ocm} = (V_{out+} + V_{out-})/2$ and compare it to a reference $V_{ref,cm} = V_{DD}/2 = 0.9\text{ V}$. The error drives the tail current source of the main OTA through a dedicated CMFB correction transistor, thereby regulating V_{ocm} . A 5T OTA is well-suited for this function because the two differential outputs of the main amplifier can be directly connected to the two inputs of the 5T CMFB OTA.

2.4 Programmable Gain Network

Programmability is achieved through a switched resistor ladder in the feedback network, realising a series-shunt feedback topology [?] as seen in 1. Digital control codes switch NMOS/PMOS transmission gates to select different R_F values from the ladder, yielding gain steps from 10 dB to 50 dB. Body-effect considerations for the MOS switches and their on-resistance contribution to gain error are discussed in section 4.

2.5 DC Offset Cancellation Loop

The DCOC architecture is a low-frequency feedback integrator [?, ?]. The differential output is sensed, low-pass filtered (integration), and fed back to the input or to a summing node at the output. The integrator time constant $\tau = R_{OC}C_{OC}$ sets the high-pass corner frequency f_{HP} , below which the closed-loop gain is suppressed. The design targets f_{HP} in the range 1–10 kHz, chosen to cancel DC and very low-frequency offsets while preserving the signal band above 10 kHz for baseband applications.

3 Theoretical Background

3.1 Bias Network (Beta-Multiplier)

The bias network uses a constant- g_m beta-multiplier to generate a stable reference current for the core OTA and CMFB loops. A PMOS mirror forces equal currents through the NMOS pair M_{14} and M_{15} , with M_{14} sized K times wider and a source degeneration resistor R_s .

$$V_{GS15} = V_{GS14} + I_{ref}R_s \quad (1)$$

Let $\beta_{15} = \mu_n C_{ox}(W/L)_{15}$ and $\beta_{14} = K\beta_{15}$. Using the I_D - V_{GS} relation for both devices gives:

$$\sqrt{\frac{2I_{ref}}{\beta_{15}}} \left(1 - \frac{1}{\sqrt{K}}\right) = I_{ref}R_s \quad (2)$$

Solving for the reference current and the resulting transconductance:

$$I_{ref} = \frac{2}{\beta_{15}R_s^2} \left(1 - \frac{1}{\sqrt{K}}\right)^2 \quad (3)$$

$$g_{m15} = \frac{2}{R_s} \left(1 - \frac{1}{\sqrt{K}}\right) \quad (4)$$

3.2 Differential-Mode and Common-Mode Gain

Based on the core amplifier schematic, the architecture is a two-stage fully differential operational amplifier. The overall differential-mode voltage gain, A_d , is the product of the gains from the first and second stages.

For the first stage (differential pair M_1, M_2 with active PMOS loads M_3, M_4), the overall differential transconductance G_{m1} is equal to the transconductance of a single input transistor, g_{m1} . To understand this, assume a perfectly matched differential pair ($g_{m1} = g_{m2}$). When a differential input voltage v_{id} is applied, it splits equally across the gate-source junctions of M_1 and M_2 as $v_{id}/2$ and $-v_{id}/2$. The resulting differential small-signal current is $i_{od} = i_{d1} - i_{d2} = g_{m1}(v_{id}/2) - g_{m2}(-v_{id}/2) = g_{m1}v_{id}$, which demonstrates that $G_{m1} = i_{od}/v_{id} = g_{m1}$.

The output resistance at the drain of M_1 is $R_{out1} = r_{o1} \parallel r_{o3}$. Therefore, the first stage differential voltage gain is:

$$A_{d1} = g_{m1}(r_{o1} \parallel r_{o3}) \quad (5)$$

For the second stage (PMOS common-source amplifiers M_5, M_6 with NMOS active loads M_7, M_8), the transconductance is $G_{m2} = g_{m5}$ and the output resistance at V_{out+} is $R_{out2} = r_{o5} \parallel r_{o7}$. The second stage gain is:

$$A_{d2} = g_{m5}(r_{o5} \parallel r_{o7}) \quad (6)$$

The total DC open-loop differential gain A_{OL} is thus:

$$A_{OL} = A_{d1} \cdot A_{d2} = g_{m1}(r_{o1} \parallel r_{o3}) \cdot g_{m5}(r_{o5} \parallel r_{o7}) \quad (7)$$

The common-mode gain is ideally zero in a balanced differential structure. However, the finite output impedance R_{SS} of the tail current source M_0 ($R_{SS} = r_{o0}$) introduces a non-zero common-mode gain in the first stage.

To derive this, we use a common-mode half-circuit analysis. When a common-mode voltage v_{ic} is applied to both inputs (M_1 and M_2), the two sides of the circuit move identically. The tail node is no longer a virtual AC ground. Instead, the tail resistance R_{SS} can be conceptually split into two parallel resistors of value $2R_{SS}$, one for each half of the circuit.

The first stage half-circuit is effectively a common-source amplifier with source degeneration $2R_{SS}$. The effective transconductance of a source-degenerated transistor is:

$$G_{m,cm} = \frac{g_{m1}}{1 + g_{m1}(2R_{SS})} \quad (8)$$

Multiplying this by the output resistance at the drain node ($r_{o1} \parallel r_{o3}$) gives the

common-mode voltage gain of the first stage:

$$A_{cm1} \approx -\frac{g_{m1}(r_{o1} \parallel r_{o3})}{1 + 2g_{m1}R_{SS}} \quad (9)$$

This common-mode signal propagates to the second stage. Because the second stage consists of two independent common-source amplifiers (M_5 and M_6) rather than a differential pair, it does not reject common-mode signals. It simply amplifies whatever signal appears at its gates by its standard single-ended gain, $A_{d2} = -g_{m5}(r_{o5} \parallel r_{o7})$.

This yields an overall common-mode gain magnitude of:

$$A_{cm} = |A_{cm1} \cdot A_{d2}| \approx \left(\frac{g_{m1}(r_{o1} \parallel r_{o3})}{1 + 2g_{m1}R_{SS}} \right) \cdot g_{m5}(r_{o5} \parallel r_{o7}) \quad (10)$$

The *Common Mode Rejection Ratio* (CMRR) is defined as the ratio of the differential gain to the common-mode gain. Since the second stage amplifies both signals equally, its gain factors out completely during the division:

$$\text{CMRR} = \left| \frac{A_d}{A_{cm}} \right| = \frac{g_{m1}(r_{o1} \parallel r_{o3}) \cdot A_{d2}}{\left(\frac{g_{m1}(r_{o1} \parallel r_{o3})}{1 + 2g_{m1}R_{SS}} \right) \cdot A_{d2}} \quad (11)$$

$$\text{CMRR} = 1 + 2g_{m1}R_{SS} \approx 2g_{m1}r_{o0} \quad (12)$$

This derivation mathematically emphasises the critical importance of a high-impedance tail current source (M_0) for achieving good common-mode rejection.

3.3 Unity-Gain Bandwidth and Phase Margin

While the core amplifier is a two-stage design, it incorporates Miller compensation networks (the capacitor C_c and nulling resistor R_c placed across the second stage) to intentionally split the poles. This forces the circuit to behave approximately as a stable, single-pole system up to its unity-gain frequency.

To derive the dominant pole, we apply Miller's theorem. The capacitor C_c bridges the input and output of the inverting second stage, which has a voltage gain of $A_{d2} = g_{m5}(r_{o5} \parallel r_{o7})$. From the perspective of the first stage's output node, this creates an enormous equivalent capacitance due to the Miller multiplication effect:

$$C_{\text{Miller}} = C_c(1 + A_{d2}) \approx C_c \cdot A_{d2} \quad (13)$$

The dominant pole f_{p1} is formed by the output resistance of the first stage ($R_{out1} =$

$r_{o1} \parallel r_{o3}$) discharging into this large Miller capacitance:

$$f_{p1} \approx \frac{1}{2\pi R_{out1} C_{Miller}} = \frac{1}{2\pi R_{out1} (C_c \cdot A_{d2})} \quad (14)$$

The unity-gain bandwidth (UGB) is defined by the DC gain-bandwidth product ($A_{OL} \cdot f_{p1}$). Since the total DC open-loop gain is $A_{OL} = A_{d1} \cdot A_{d2} = (g_{m1} R_{out1}) \cdot A_{d2}$, we can substitute these values to find the UGB:

$$\text{UGB} \approx A_{OL} \cdot f_{p1} = (g_{m1} R_{out1} A_{d2}) \cdot \frac{1}{2\pi R_{out1} A_{d2} C_c} \quad (15)$$

$$\text{UGB} \approx \frac{g_{m1}}{2\pi C_c} \quad (16)$$

Crucially, for a Miller-compensated two-stage amplifier, the UGB is dictated by the compensation capacitor C_c and the first-stage transconductance g_{m1} , completely independent of the load capacitance C_L .

To understand the location of the non-dominant pole f_{p2} , we must analyse the output node of the second stage at high frequencies. At frequencies beyond the dominant pole, the compensation capacitor C_c acts almost as an AC short circuit between the gate and drain of the second-stage amplifier M_5 (or M_6). This high-frequency short effectively diode-connects M_5 , drastically reducing the resistance seen at the output node from the DC value of $r_{o5} \parallel r_{o7}$ down to approximately $1/g_{m5}$.

The capacitance at this final output node is dominated by the load capacitance C_L . Therefore, the non-dominant pole, formed by this reduced high-frequency output resistance discharging into C_L , is approximated by:

$$f_{p2} \approx \frac{1}{2\pi \cdot (1/g_{m5}) \cdot C_L} = \frac{g_{m5}}{2\pi C_L} \quad (17)$$

In a standard Miller-compensated amplifier, the feedforward signal path - where high-frequency signals bypass the inverting second stage completely and travel directly through C_c to the output - creates a Right-Half-Plane (RHP) zero. This RHP zero is notorious because it flattens the gain magnitude curve while simultaneously degrading the phase margin.

Let v_x be the small-signal voltage at the gate of the second stage transistor M_5 , and v_{out} be the voltage at its drain. The small-signal current leaving the output node through the transconductor M_5 is $i_{gm} = g_{m5} v_x$. The compensation network (C_c and R_c in series) provides a parallel feedforward path. The current entering the output node from this feedforward path is $i_{ff} = (v_x - v_{out}) / (R_c + 1/sC_c)$.

At the zero frequency, we set $v_{out} = 0$. By Kirchhoff's Current Law (KCL), the sum of

currents entering and leaving the node must balance:

$$i_{ff} = i_{gm} \quad (18)$$

$$\frac{v_x - 0}{R_c + \frac{1}{sC_c}} = g_{m5}v_x \quad (19)$$

Assuming $v_x \neq 0$, we can cancel it from both sides and solve for s :

$$\frac{sC_c}{1 + sR_cC_c} = g_{m5} \quad (20)$$

$$sC_c = g_{m5} + s \cdot g_{m5}R_cC_c \quad (21)$$

$$s(C_c - g_{m5}R_cC_c) = g_{m5} \quad (22)$$

$$s_z = \frac{g_{m5}}{C_c(1 - g_{m5}R_c)} = \frac{1}{C_c(1/g_{m5} - R_c)} \quad (23)$$

Converting from the complex variable s to frequency $f = s/2\pi$, we get our expression for the zero:

$$f_z = \frac{1}{2\pi C_c(1/g_{m5} - R_c)} \quad (24)$$

If the nulling resistor is absent ($R_c = 0$), the zero sits at $s = +g_{m5}/C_c$. Because the root is positive, it lies in the Right-Half-Plane (RHP).

Equating the new LHP zero frequency to the non-dominant pole frequency:

$$\frac{1}{2\pi C_c(R_c - 1/g_{m5})} = \frac{g_{m5}}{2\pi C_L} \quad (25)$$

Solving for R_c yields the exact resistance required to perfectly cancel the non-dominant pole, theoretically extending the -20 dB/decade roll-off and significantly improving phase margin:

$$R_c = \frac{1}{g_{m5}} \left(1 + \frac{C_L}{C_c} \right) \quad (26)$$

3.4 Slew Rate

The slew rate (SR) is limited by the maximum rate at which the load capacitor can be charged:

$$\text{SR} = \frac{I_{tail}}{C_L}, \quad (27)$$

where I_{tail} is the tail bias current of the differential pair.

3.5 Noise Analysis

The input-referred noise voltage spectral density of a single MOSFET consists of thermal and flicker ($1/f$) components:

$$\overline{v_n^2}(f) = \frac{4kT\gamma}{g_m} + \frac{K_f}{C_{ox}WLf} \quad (28)$$

The circuit is a single-stage fully differential cascode OTA. The tail current source M_0 generates only common-mode noise, which is strongly attenuated by the CMRR and is therefore negligible. The dominant input-referred noise contributions come from the four active branches: the input pair (M_1, M_2), the PMOS loads (M_3, M_4), and the upper and lower cascode devices (M_5, M_6 and M_7, M_8 , respectively).

The total output noise current (referred to one output node, then doubled for full differential) is:

$$\overline{i_{n,out}^2} = 2 \left(g_{m1}^2 \overline{v_{n1}^2} + g_{m3}^2 \overline{v_{n3}^2} + g_{m5}^2 \overline{v_{n5}^2} + g_{m7}^2 \overline{v_{n7}^2} \right) \quad (29)$$

Cascode transistors (M_5 – M_8) contribute noise currents of order g_{m5}/g_{m1} relative to the input pair. Since cascode devices are typically biased at lower g_m than the input pair, their contribution is small but not negligible. Dividing by g_{m1}^2 to refer back to the differential input:

$$\overline{v_{n,in}^2} = 2 \left[\overline{v_{n1}^2} + \left(\frac{g_{m3}}{g_{m1}} \right)^2 \overline{v_{n3}^2} + \left(\frac{g_{m5}}{g_{m1}} \right)^2 \overline{v_{n5}^2} + \left(\frac{g_{m7}}{g_{m1}} \right)^2 \overline{v_{n7}^2} \right] \quad (30)$$

Substituting Equation 28 and separating into thermal and flicker components:

Total Thermal Noise:

$$\overline{v_{n,in(th)}^2} = \frac{8kT\gamma}{g_{m1}} \left[1 + \frac{g_{m3}}{g_{m1}} + \frac{g_{m5}}{g_{m1}} + \frac{g_{m7}}{g_{m1}} \right] \quad (31)$$

Total Flicker ($1/f$) Noise:

$$\overline{v_{n,in(1/f)}^2} = \frac{2}{f C_{ox}} \left[\frac{K_{fN}}{W_1 L_1} + \left(\frac{g_{m3}}{g_{m1}} \right)^2 \frac{K_{fP}}{W_3 L_3} + \left(\frac{g_{m5}}{g_{m1}} \right)^2 \frac{K_{fP}}{W_5 L_5} + \left(\frac{g_{m7}}{g_{m1}} \right)^2 \frac{K_{fN}}{W_7 L_7} \right] \quad (32)$$

The total input-referred noise is $\overline{v_{n,in}^2} = \overline{v_{n,in(th)}^2} + \overline{v_{n,in(1/f)}^2}$. To minimise noise, g_{m1} should be maximised, and the cascode transistors should be sized such that their g_m ratios to g_{m1} remain small.

3.6 CMFB Loop Gain Derivation

The Common-Mode Feedback (CMFB) loop regulates the DC output voltage of the fully differential amplifier. In this two-stage architecture, the loop encompasses the CMFB

sensing network, the CMFB error amplifier, and both stages of the core amplifier.

To derive the CMFB loop gain (T_{cm}), we conceptually break the closed loop and trace a common-mode test voltage (V_{test}) through the system:

1. **Sensing Factor (A_{sense}):** The resistive divider extracts the common-mode voltage from the outputs. For a pure common-mode signal, the voltage at the center tap tracks the output exactly, yielding a sensing gain of unity: $A_{sense} = 1$.
2. **CMFB Amplifier Gain (A_{cmfb}):** The 5T CMFB OTA compares V_{sense} to a static V_{ref} . Because V_{ref} is an AC ground, the differential pair (M10, M11) is driven single-endedly, meaning only half the input AC voltage drops across the active transistor M10. The active loads (M12, M13) are diode-connected, presenting an impedance of roughly $1/g_{m12}$. The resulting voltage gain is:

$$A_{cmfb} = -\frac{g_{m10}}{2g_{m12}} \quad (33)$$

3. **Core Amplifier First Stage Gain ($A_{1,cm}$):** The control signal V_{cmfb} drives the gates of the PMOS active loads (M3, M4), which act as common-source amplifiers. The common-mode output resistance $R_{out1,cm}$ is the parallel combination of the resistance looking up into M3 (r_{o3}) and the resistance looking down into M1 degenerated by the tail current source M0 (r_{o0}):

$$R_{out1,cm} = r_{o3} \parallel [r_{o1} + (1 + g_{m1}r_{o1})(2r_{o0})] \approx r_{o3} \quad (34)$$

The common-mode gain for this stage is $A_{1,cm} = -g_{m3}R_{out1,cm}$.

4. **Core Amplifier Second Stage Gain ($A_{2,cm}$):** The output of the first stage drives the PMOS common-source amplifiers of the second stage (M5, M6). The output resistance is simply the parallel combination of the driver (M5) and the NMOS current source load (M7):

$$R_{out2,cm} = r_{o5} \parallel r_{o7} \quad (35)$$

The common-mode gain for this stage is $A_{2,cm} = -g_{m5}R_{out2,cm}$.

Combining these cascaded stages, the total low-frequency CMFB loop gain is the product of the individual stage gains. The three signal inversions result in a net negative value, ensuring negative feedback:

$$T_{cm} = A_{sense} \cdot |A_{cmfb}| \cdot |A_{1,cm}| \cdot |A_{2,cm}| = \frac{g_{m10}}{2g_{m12}} \cdot g_{m3}R_{out1,cm} \cdot g_{m5}R_{out2,cm} \quad (36)$$

To meet the project target of $T_{cm} \geq 40$ dB (a linear magnitude of 100 V/V), the

transconductances and output resistances must be sized to satisfy:

$$\frac{g_{m10} \cdot g_{m3} \cdot g_{m5}}{2g_{m12}} \cdot R_{out1,cm} \cdot R_{out2,cm} \geq 100 \quad (37)$$

3.7 DC Offset Cancellation Principle

The DCOC loop places an integrator in a feedback path from the output of the PGA back to a summing node at the input (or internal node). For DC signals, the integrator gain is infinite, driving the output DC offset to zero. The closed-loop transfer function of the PGA with DCOC is a high-pass function:

$$H_{DCOC}(s) = A_{CL} \cdot \frac{s/\omega_{HP}}{1 + s/\omega_{HP}}, \quad (38)$$

where $\omega_{HP} = 1/(R_{OC}C_{OC})$ is the high-pass pole. For f_{HP} between 1 kHz and 10 kHz, the integrator time constant must satisfy

$$\tau_{OC} = \frac{1}{2\pi f_{HP}}. \quad (39)$$

4 Hand Calculations and Transistor Sizing

This section summarizes the final sizing calculations used for the implemented design. The closed-loop bandwidth requirement at 50 dB gain sets the minimum UGB:

$$UGB \geq 316.2 \times 750 \text{ kHz} \approx 237 \text{ MHz} \quad (40)$$

The total current budget for the OTA and CMFB is constrained by the 1 mW power target at 1.8 V:

$$I_{tot,max} = \frac{1 \text{ mW}}{1.8 \text{ V}} \approx 555.5 \text{ } \mu\text{A} \quad (41)$$

4.1 Bias Network Calculations

We target $I_{ref} = 10 \text{ } \mu\text{A}$ with $K = 4$ and $V_{ov15} \approx 0.2 \text{ V}$. The reference transconductance and source degeneration resistance are:

$$g_{m15} = \frac{2I_{ref}}{V_{ov15}} = 100 \text{ } \mu\text{A/V} \quad (42)$$

$$R_s = \frac{2}{g_{m15}} \left(1 - \frac{1}{\sqrt{K}} \right) = 10 \text{ k}\Omega \quad (43)$$

Assume $\mu_n C_{ox} \approx 270 \text{ } \mu\text{A/V}^2$, $\mu_p C_{ox} \approx 70 \text{ } \mu\text{A/V}^2$, and $L = 1.0 \text{ } \mu\text{m}$ unless noted. The resulting device sizes are:

- **NMOS core pair** (M_{15}, M_{14}):

$$(W/L)_{15} = \frac{2I_{ref}}{\mu_n C_{ox} V_{ov15}^2} \approx 1.85 \Rightarrow W_{15} = 1.85 \mu\text{m}$$

$$W_{14} = 4W_{15} = 7.4 \mu\text{m}$$

- **PMOS mirror** (M_{16}, M_{17}):

$$(W/L)_{16,17} = \frac{2I_{ref}}{\mu_p C_{ox} V_{ov,p}^2} \approx 7.14 \Rightarrow W_{16,17} = 7.15 \mu\text{m}$$

- **Output bias branch** (M_{19}, M_{20}, M_{21}): We want a stable output bias node voltage of 0.65 V and that will be used to bias all the tail transistors. If we inject the same current into this branch, we can perform the following calculations. To limit the CTAT injection while keeping the NMOS bias node V_b low, M_{19} is intentionally weakened and M_{20} is strengthened. The NMOS sink ratio sets a lower V_b while preserving a smaller bias branch for mirroring.

We want it to inject $10\mu\text{A}$:

$$10 = \frac{1}{2}(40\mu\text{A}/\text{V}^2) \left(\frac{W}{L}\right)_{19} (0.7\text{V})^2$$

$$10 = 20 \times 0.49 \times \left(\frac{W}{L}\right)_{19} \Rightarrow \left(\frac{W}{L}\right)_{19} \approx 1$$

M_{20} 's gate is also tied to M_{15} , so $V_{G20} = 0.65\text{V}$. For M_{20} to allow $10\mu\text{A}$ through, it needs an overdrive voltage. Let's design it to have $V_{ov20} = 0.1\text{V}$.

$$V_{GS20} = V_{th} + V_{ov20} = 0.45\text{V} + 0.1\text{V} = 0.55\text{V}$$

Because $V_{GS20} = V_G - V_{Source}$, the voltage at the source of M_{20} (which is the drain of M_{21}) must be exactly 0.1V ($0.65\text{V} - 0.55\text{V}$).

$$10\mu\text{A} = \frac{1}{2}(170\mu\text{A}/\text{V}^2) \left(\frac{W}{L}\right)_{20} (0.1\text{V})^2$$

$$10 = 0.85 \times \left(\frac{W}{L}\right)_{20} \Rightarrow \left(\frac{W}{L}\right)_{20} \approx 12$$

M_{21} 's Gate is $V_b = 0.65\text{V}$. (So $V_{ov21} = 0.2\text{V}$). M_{21} 's Drain is 0.1V (calculated above). Because $V_{DS}(0.1\text{V}) < V_{ov}(0.2\text{V})$, M_{21} is mathematically forced into the

Triode (Linear) region! We must use the triode equation to size it for $10\mu\text{A}$:

$$\begin{aligned}
 I_D &= \mu_n C_{ox} \left(\frac{W}{L} \right) \left[(V_{GS} - V_{th})V_{DS} - \frac{1}{2}V_{DS}^2 \right] \\
 10\mu\text{A} &= 170\mu\text{A}/\text{V}^2 \times \left(\frac{W}{L} \right)_{21} \times [(0.2\text{V})(0.1\text{V}) - 0.5(0.1\text{V})^2] \\
 10 &= 170 \times \left(\frac{W}{L} \right)_{21} \times [0.02 - 0.005] \\
 10 &= 170 \times 0.015 \times \left(\frac{W}{L} \right)_{21} = 2.55 \times \left(\frac{W}{L} \right)_{21} \\
 \left(\frac{W}{L} \right)_{21} &\approx 4
 \end{aligned}$$

$$W_{19} = 0.5 \mu\text{m}, L_{19} = 0.5 \mu\text{m}, \quad W_{20} = 10 \mu\text{m}, L_{20} = 0.5 \mu\text{m}, \quad W_{21} = 2 \mu\text{m}, L_{21} = 0.5 \mu\text{m}$$

- **Start-up device (M_{18}):** Since M_{18} is just a kickstarter that should turn off immediately, I make it as weak as possible so it doesn't leak current during normal operation.

$$W_{18} = 1.0 \mu\text{m}, \quad L_{18} = 4.0 \mu\text{m}$$

4.2 Core Differential Amplifier (First Stage)

The core amplifier sizing begins with speed, power, and stability constraints. While the closed-loop specification dictates a minimum unity-gain bandwidth of 250 MHz, we target a nominal UGB of ≈ 310 MHz to ensure robust margins across process, voltage, and temperature (PVT) variations:

$$UGB_{nom} \approx 310 \text{ MHz} \tag{44}$$

From the UGB relation, the required transconductance-to-compensation ratio is:

$$\frac{g_{m1}}{C_c} = 2\pi \cdot UGB_{nom} \approx 1.95 \times 10^9 \text{ s}^{-1} \tag{45}$$

To achieve a phase margin of $\approx 60^\circ$, we carefully place the non-dominant pole (ω_{p2}). Keeping the phase at ω_{UGB} near -120° requires:

$$\frac{\omega_{UGB}}{\omega_{p2}} \approx \tan(30^\circ) \implies \omega_{p2} \approx 1.73 \omega_{UGB} \tag{46}$$

With a relaxed ratio of $\omega_{p2}/\omega_{UGB} \approx 1.59$ to account for nulling-resistor zero placement,

and given $\omega_{p2} \approx g_{m5}/C_L$, we obtain:

$$g_{m5} = 1.59 \frac{g_{m1}}{C_c} C_L \approx 3.11 \text{ mA/V} \quad (C_L = 1 \text{ pF}) \quad (47)$$

Choosing a compensation capacitor of $C_c \approx 0.4 \text{ pF}$, the required first-stage transconductance is:

$$g_{m1} = 2\pi UGB_{nom} \cdot C_c \approx 0.78 \text{ mA/V} \quad (48)$$

Instead of targeting a specific transconductance efficiency ratio, we directly allocate a current budget to the first stage that satisfies our power constraints while allowing sufficient bandwidth. We set the tail current to $75 \mu\text{A}$:

$$I_{tail,main} = 75 \mu\text{A} \implies I_{D1,2} = 37.5 \mu\text{A} \quad (49)$$

To minimize parasitic capacitance at the intermediate nodes and push higher-order poles to safely high frequencies, the core device lengths are kept to a minimum (e.g., $L_{1-4} = 0.5 \mu\text{m}$).

Assuming nominal process parameters of $\mu_n C_{ox} \approx 280 \mu\text{A/V}^2$ and $\mu_p C_{ox} \approx 70 \mu\text{A/V}^2$, the required width for the NMOS input pair can be explicitly calculated using the square-law relation $g_m = \sqrt{2\mu C_{ox}(W/L)I_D}$:

$$W_{1,2} = \frac{g_{m1}^2 L_{1,2}}{2\mu_n C_{ox} I_{D1}} = \frac{(0.78 \text{ mA/V})^2 \times 0.5 \mu\text{m}}{2 \times 280 \mu\text{A/V}^2 \times 37.5 \mu\text{A}} \approx 14.5 \mu\text{m} \quad (50)$$

For the PMOS active load ($M_{3,4}$), establishing a target overdrive voltage of $V_{ov3} \approx 0.13 \text{ V}$ yields:

$$W_{3,4} = \frac{2I_{D3}L_{3,4}}{\mu_p C_{ox} V_{ov3}^2} = \frac{2 \times 37.5 \mu\text{A} \times 0.5 \mu\text{m}}{70 \mu\text{A/V}^2 \times (0.13 \text{ V})^2} \approx 31.7 \mu\text{m} \quad (51)$$

Similarly, the NMOS tail current source (M_0) supporting $75 \mu\text{A}$ with $V_{ov0} \approx 0.13 \text{ V}$ requires:

$$W_0 = \frac{2I_{tail}L_0}{\mu_n C_{ox} V_{ov0}^2} = \frac{2 \times 75 \mu\text{A} \times 0.5 \mu\text{m}}{280 \mu\text{A/V}^2 \times (0.13 \text{ V})^2} \approx 15.8 \mu\text{m} \quad (52)$$

4.3 Second-Stage Common-Source Amplifier

The second stage must drive the load and push the non-dominant pole outwards. To achieve the rigorous $g_{m5} \approx 3.11 \text{ mA/V}$ requirement, we directly bias the second stage with a high current:

$$I_{D5,6} = 200 \mu\text{A} \quad (53)$$

Because of this large bias current, short-channel devices would suffer from severe channel-length modulation, drastically degrading output resistance. To carry the bulk

of the overall 76 dB open-loop gain requirement, we increase the channel length of the second stage to $L_{5-8} = 1.0 \mu\text{m}$.

To maintain the critical 3.11 mA/V transconductance with these longer channels, the width for the PMOS driver (M_5, M_6) must be scaled up significantly:

$$W_{5,6} = \frac{g_{m5}^2 L_{5,6}}{2\mu_p C_{ox} I_{D5}} = \frac{(3.11 \text{ mA/V})^2 \times 1.0 \mu\text{m}}{2 \times 70 \mu\text{A/V}^2 \times 200 \mu\text{A}} \approx 345 \mu\text{m} \quad (54)$$

Due to its large size, this device is implemented with a multi-finger layout (e.g., $m = 20$, $W_{unit} \approx 17.25 \mu\text{m}$) to minimize gate resistance.

The NMOS current source loads (M_7, M_8) mirror the 200 μA current. Targeting a slightly higher headroom $V_{ov7} \approx 0.15 \text{ V}$:

$$W_{7,8} = \frac{2I_{D7}L_{7,8}}{\mu_n C_{ox} V_{ov7}^2} = \frac{2 \times 200 \mu\text{A} \times 1.0 \mu\text{m}}{280 \mu\text{A/V}^2 \times (0.15 \text{ V})^2} \approx 63.5 \mu\text{m} \quad (55)$$

The core OTA current consumption totals $I_{OTA} \approx 75 \mu\text{A} + 2 \times 200 \mu\text{A} = 475 \mu\text{A}$. Including the CMFB network, the total current is approximately 535 μA (roughly 0.96 mW at 1.8 V).

4.4 Miller Compensation Network (C_c, R_c)

The unity-gain bandwidth and non-dominant pole are established as:

$$UGB \approx \frac{g_{m1}}{2\pi C_c} \approx 310 \text{ MHz} \quad (56)$$

$$\omega_{p2} \approx \frac{g_{m5}}{C_L} \approx 3.11 \times 10^9 \text{ rad/s} \quad (57)$$

To cancel the phase degradation from the non-dominant pole, we utilize a nulling resistor (R_c) to place the Left-Half-Plane (LHP) zero directly on top of ω_{p2} . The zero frequency is:

$$\omega_z = \frac{1}{C_c \left(R_c - \frac{1}{g_{m5}} \right)} \quad (58)$$

Equating $\omega_z = \omega_{p2}$ and solving for R_c yields:

$$R_c = \frac{1}{g_{m5}} \left(1 + \frac{C_L}{C_c} \right) \quad (59)$$

Substituting $C_L = 1 \text{ pF}$, $C_c = 0.4 \text{ pF}$, and $g_{m5} = 3.11 \text{ mA/V}$:

$$R_c = \frac{1}{3.11 \times 10^{-3}} (1 + 2.5) \approx 1.12 \text{ k}\Omega \quad (60)$$

4.5 Common-Mode Feedback (CMFB) Circuit Design

We target a tail current of $10\ \mu\text{A}$. Because the gate of M9 connects directly to the same temperature-compensated bias node as the core OTA's tail, we can scale its dimensions relative to the core's tail current source (M0). To sink exactly a current of $10\ \mu\text{A}$, $(\frac{W}{L})_9 = \frac{8}{4} = 2$. Using a standard matching length of $L_9 = 0.5\ \mu\text{m}$, the width is chosen as $W_9 = 1.0\ \mu\text{m}$.

For a balanced common-mode detection, the $10\ \mu\text{A}$ tail current splits evenly, meaning M10 and M11 each carry $5\ \mu\text{A}$. The gate of M11 is driven by the ideal fixed reference $V_{ref} = 0.9\ \text{V}$, while the gate of M10 connects to the resistor sensing loop (which also ideally rests at $0.9\ \text{V}$). We design the input pair for an overdrive voltage of $V_{ov} = 0.12\ \text{V}$ to provide a healthy transconductance (g_m).

$$I_D = \frac{1}{2}\mu_n C_{ox} \left(\frac{W}{L}\right)_{10,11} V_{ov}^2 \quad (61)$$

$$5\ \mu\text{A} = \frac{1}{2}(270\ \mu\text{A}/\text{V}^2) \left(\frac{W}{L}\right)_{10,11} (0.12\ \text{V})^2 \quad (62)$$

$$\left(\frac{W}{L}\right)_{10,11} = \frac{5}{1.91} \approx 3 \quad (63)$$

Selecting $L_{10,11} = 0.5\ \mu\text{m}$ gives $W_{10,11} = 1.5\ \mu\text{m}$.

These diode-connected transistors must pull exactly $5\ \mu\text{A}$ up to V_{DD} ($1.8\ \text{V}$).

Plugging an overdrive of $0.14\ \text{V}$ into the PMOS current equation:

$$5\ \mu\text{A} = \frac{1}{2}(70\ \mu\text{A}/\text{V}^2) \left(\frac{W}{L}\right)_{12,13} (0.14\ \text{V})^2 \quad (64)$$

$$\left(\frac{W}{L}\right)_{12,13} = \frac{5}{0.644} \approx 8 \quad (65)$$

Choosing length $L_{12,13} = 0.5\ \mu\text{m}$ gives $W_{12,13} = 4.0\ \mu\text{m}$.

Two large resistors are placed between V_{out+} and V_{out-} to extract their average (common-mode) voltage. If these resistors are too small in value, they will load the amplifier and heavily degrade the output resistance, ruining the differential gain. To preserve high output resistance, I use two **1 M Ω** resistors.

4.6 Programmable Gain Network Design

To allow the full analog voltage signal to pass without clipping, we use a CMOS Transmission Gate (T-Gate), which places an NMOS and a PMOS in parallel.

The ON-resistance (R_{on}) of a MOSFET in the deep linear (triode) region is determined by the overdrive voltage and physical dimensions:

- **NMOS ON-Resistance:** $R_{on,N} = \frac{1}{\mu_n C_{ox} \left(\frac{W}{L}\right)_N (V_{DD} - V_{in} - V_{th,N})}$
- **PMOS ON-Resistance:** $R_{on,P} = \frac{1}{\mu_p C_{ox} \left(\frac{W}{L}\right)_P (V_{in} - |V_{th,P}|)}$

The total equivalent resistance of the switch is the parallel combination:

$$R_{on,TG} = R_{on,N} \parallel R_{on,P} \quad (66)$$

As the analog input signal (V_{in}) swings up and down, $R_{on,N}$ and $R_{on,P}$ change dynamically. However, when placed in parallel, their changes counteract each other. This results in a relatively constant, flat R_{on} across the entire voltage swing, which prevents Total Harmonic Distortion (THD) in the amplifier.

We size the switches heavily ($W_N = 10 \mu\text{m}$, $W_P = 30 \mu\text{m}$, $L = 0.18 \mu\text{m}$) to achieve a nominal flat resistance. When a switch closes, it connects a specific node in the resistor string directly to the output. This places R_{on} strictly in series with the feedback resistors.

We establish the input resistor as $R_{in} = 5.0 \text{ k}\Omega$.

- **10 dB** (3.162 V/V): $R_{F1} = 15.8 \text{ k}\Omega$
- **20 dB** (10 V/V): $R_{F2} = 34.2 \text{ k}\Omega$
- **40 dB** (100.0 V/V): $R_{F3} = 450.00 \text{ k}\Omega$
- **50 dB** (316.228 V/V): $R_{F4} = 1.08 \text{ M}\Omega$

5 Simulation Setup and Results

All simulations are performed in Cadence Virtuoso Spectre using the SCL 180 nm CMOS process models (TT corner, $T = 27^\circ\text{C}$, $V_{DD} = 1.8 \text{ V}$ unless otherwise stated). A load capacitance of $C_L = 1 \text{ pF}$ is connected from each differential output to ground.

5.1 Schematics

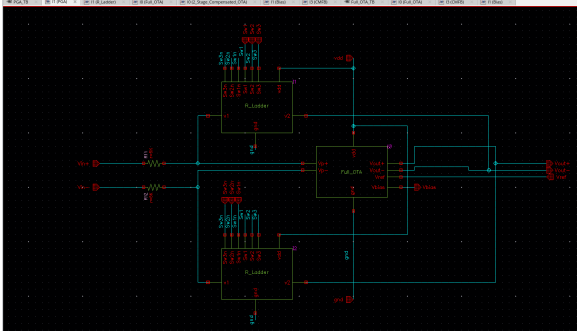


Figure 2: Overall PGA circuit with resistive ladder and amplifier

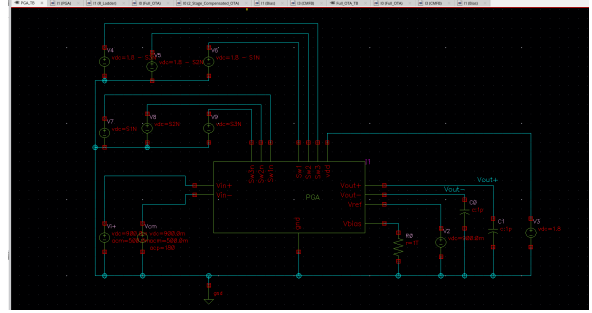


Figure 3: Test bench to test PGA. The voltage sources give the code input to the resistive ladder.

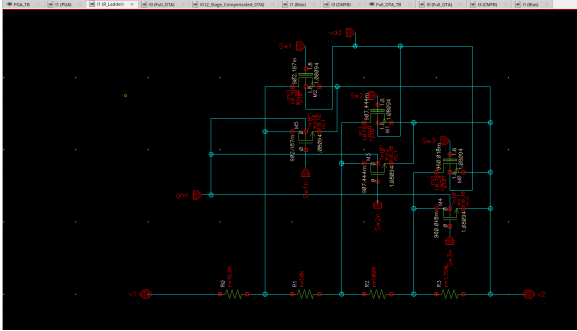


Figure 4: Resistive ladder circuit, made with transmission gates

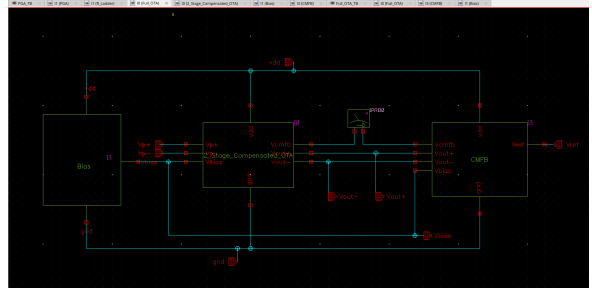


Figure 5: Sub circuits in the Full OTA cell - Bias circuit, 2 stage amplifier and the CMFB circuit. We can see the break-point at which the probe is inserted as well.

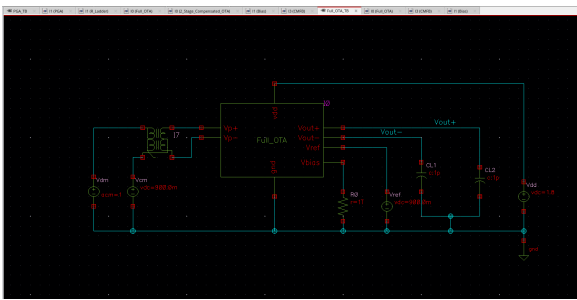


Figure 6: Test bench to test OTA in open loop configuration

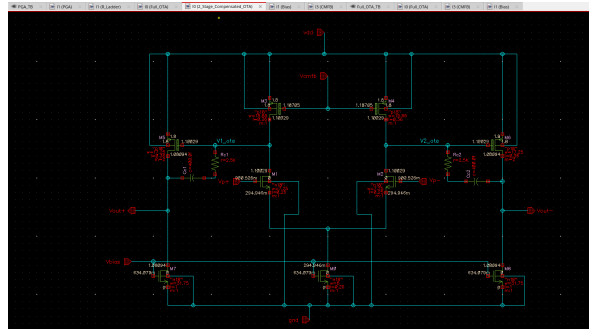


Figure 7: 2 stage Miller compensated amplifier circuit

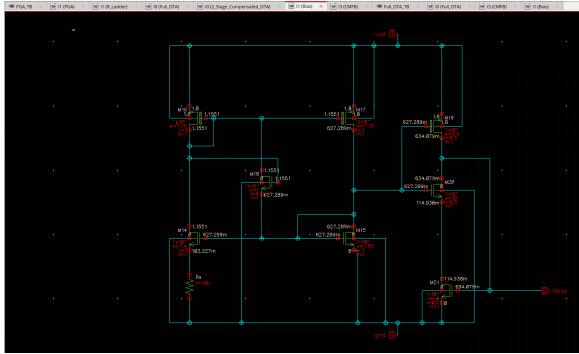


Figure 8: Bias circuit that generates the voltage for the tail transistors and active loads

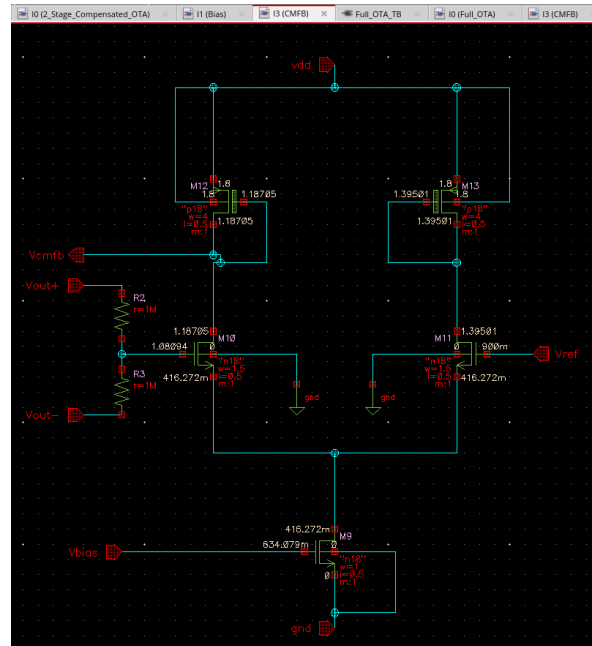


Figure 9: CMFB circuit - a differential amplifier with diode connected load

5.2 DC Operating Point

Table 1: DC Operating Point: Node Voltages and Branch Currents (TT Corner, 27°C)

Node / Branch	Expected (V or μA)	Simulated	Unit
<i>Supply and Common-Mode Nodes</i>			
V_{DD}	1.8	1.8	V
V_{out+}	≈ 0.9	1.08	V
V_{out-}	≈ 0.9	1.08	V
V_{bias}	—	634.079	mV
V_{cmfb}	—	1.18705	V
V_{ref} (CMFB reference)	—	900	mV
<i>Core OTA — Differential Pair</i>			
$I_{D,M1} = I_{D,M2}$	37.5	48.163	μA
$g_{m,M1} = g_{m,M2}$	780	708.363	$\mu\text{A}/\text{V}$
$g_{m,M3} = g_{m,M4}$	—	0.46	mA/V
<i>Core OTA — PMOS Amplifier ($M5, M6$)</i>			
$I_{D,M5} = I_{D,M6}$	200	209.4	μA
$g_{m,M5} = g_{m,M6}$	3.11	1.8	mA/V
<i>Core OTA — NMOS Tail Mirrors ($M7, M8$ and $M0$)</i>			
$I_{D,M7}$	200	209.364	μA
$g_{m,M7}$	—	1802.38	$\mu\text{A}/\text{V}$
$I_{D,M8}$	200	209.364	μA
$I_{D,M0}$	75	96.132	μA
$g_{m,M0}$	—	927.644	$\mu\text{A}/\text{V}$
<i>CMFB Circuit (5T-OTA based)</i>			
$I_{D,M10}$	5	8.07505	μA
$g_{m,M10}$	—	98.1691	$\mu\text{A}/\text{V}$
$I_{D,M9}$ (CMFB tail bias)	10	8.4190	μA
$g_{m,M9}$	—	78.472	$\mu\text{A}/\text{V}$
<i>Bias Circuit (Current Reference)</i>			
$I_{D,M16} = I_{D,M17}$ (PMOS mirror)	10	10.4655	μA
$I_{D,M14}$	10	10.3227	μA
$I_{D,M21}$	10	13.7473	μA
V_{bias} (output of bias cell)	650	634.079	mV
<i>Total Power</i>			
$I_{DD,\text{total}}$	< 556	≈ 558.3	μA
P_{total}	< 1.0	≈ 1	mW

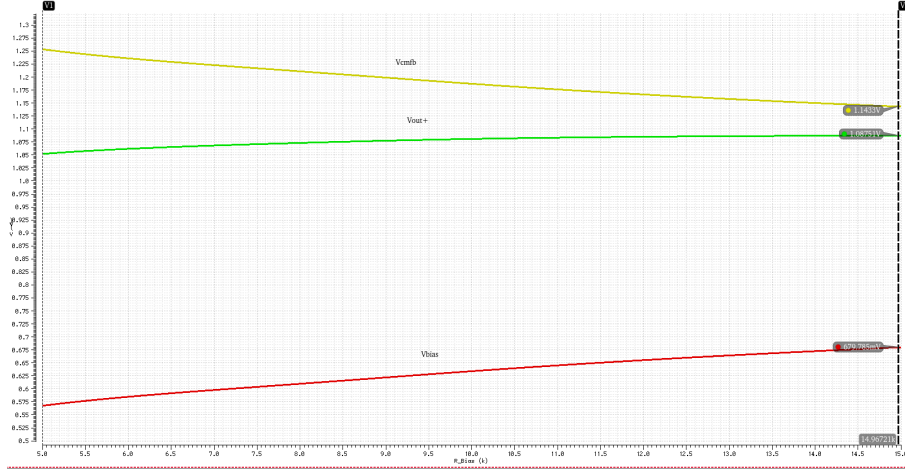


Figure 10: Sweeping the bias resistor from 5K to 15K (original value 10K, so 50% variation), the output node varies by 3.4%, V_{cmfb} by 9.26%, and the tail-bias voltage by 17.7%.

5.3 AC / Small-Signal Analysis

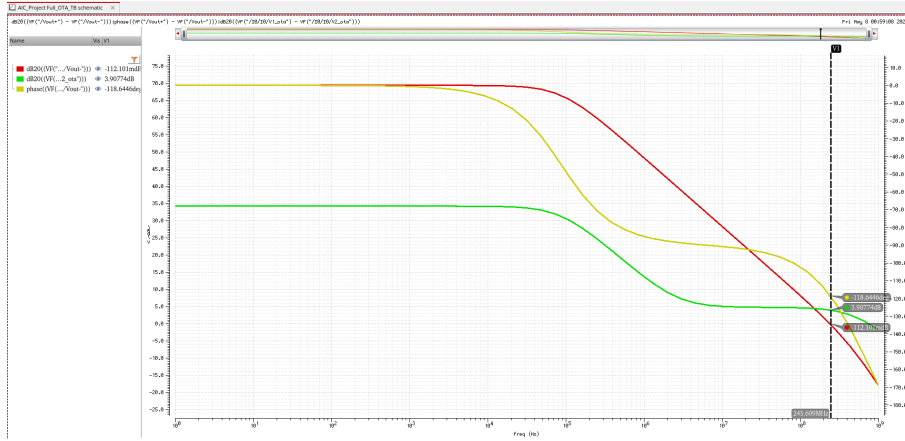
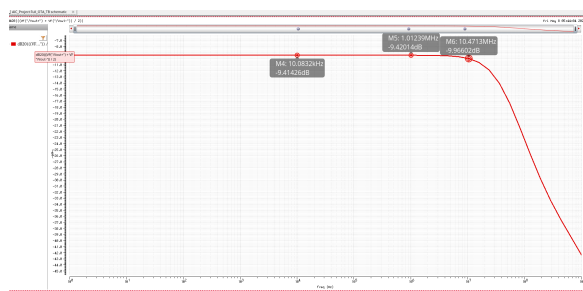
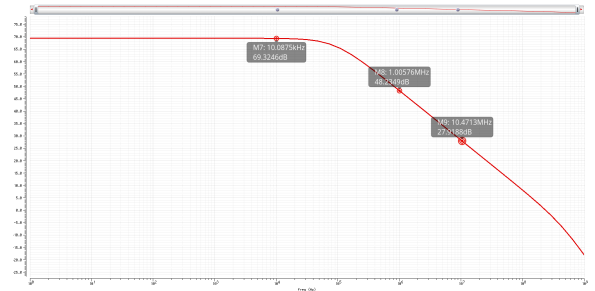


Figure 11: Frequency analysis of the OTA block. The observed unity-gain bandwidth is 245.6 MHz, the open-loop gain is 69.4 dB, the open-loop phase margin is 61.4°, and the -3 dB frequency is 89 kHz, all meeting the specifications. The first stage drives 34.2 dB, while the second stage drives 25 dB.



(a) Common-mode gain A_{cm} .



(b) Differential-mode gain A_{dm} .

Figure 12: Common-mode and differential-mode gain responses of the OTA.

Table 2: CMRR extracted from AC analysis

Frequency	A_{cm} (dB)	A_{dm} (dB)	CMRR (dB)
10 kHz	-9.41	69.32	78.74
1 MHz	-9.42	48.23	57.65
10 MHz	-9.97	27.92	37.89

CMRR values are computed as $A_{dm} - A_{cm}$ in dB.

5.4 Stability Analysis



Figure 13: The CMFB loop phase margin is 72°, the bandwidth is 21.7 MHz, and the loop gain is 55.56 dB. The bandwidth is less than 10x the UGB and does not overcompensate small high-frequency signals.

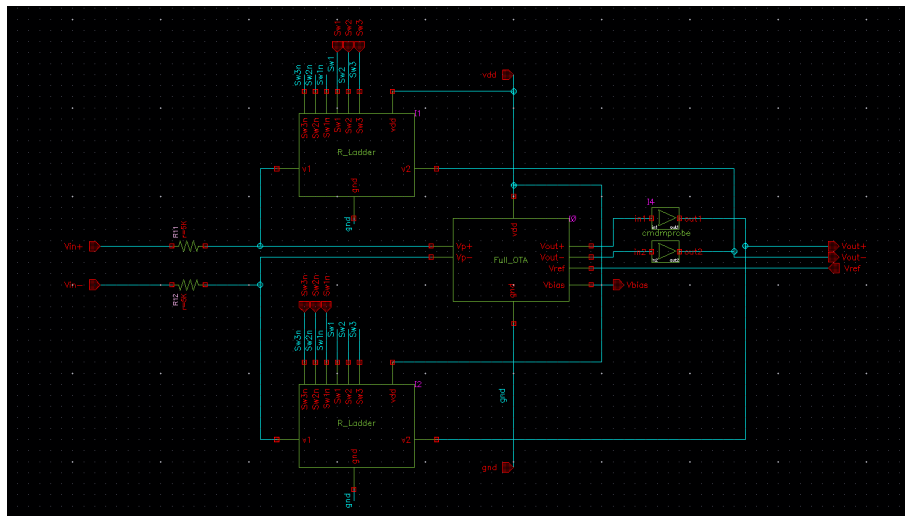


Figure 14: Breaking the PGA loop with the cmdmprobe

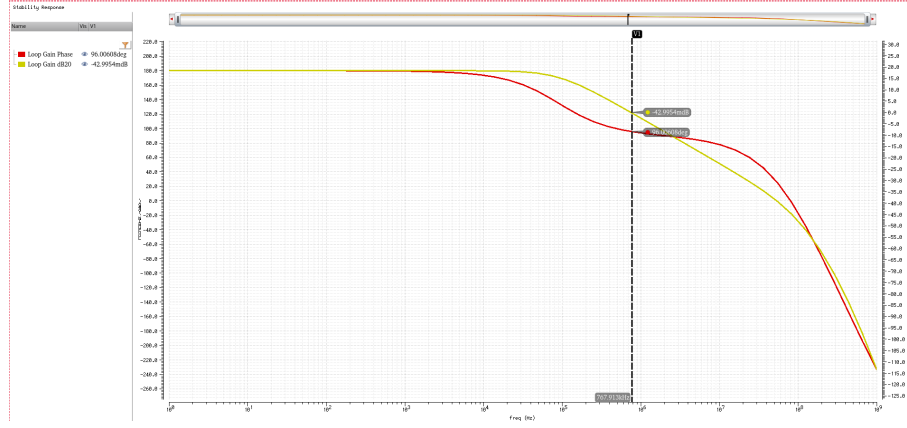


Figure 15: Loop phase and Loop gain plot for the PGA at a specific gain.

Table 3: Loop gain performance across gain settings

Gain Setting	Gain (dB)	Bandwidth	Phase Margin	Gain Margin
10 dB	47.22 dB	59.1 MHz	80°	29.47 dB
20 dB	42.21 dB	17.9 MHz	83.4°	29.31 dB
40 dB	27.59 dB	2.27 MHz	90.9°	41.51 dB
50 dB	18.44 dB	750.4 kHz	95.9°	44.41 dB

5.5 Transient and Large-Signal Analysis

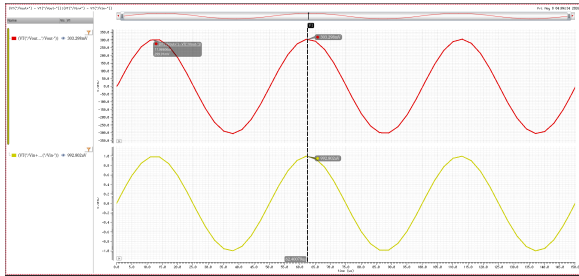
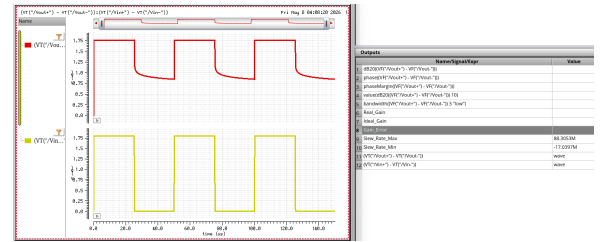
(a) Sine-wave transient response (gain ≈ 305.5).(b) Step response used to estimate slew rate (88.3/17 V/ μ s).

Figure 16: Large-signal transient response of the PGA at maximum gain.

5.6 Noise Analysis

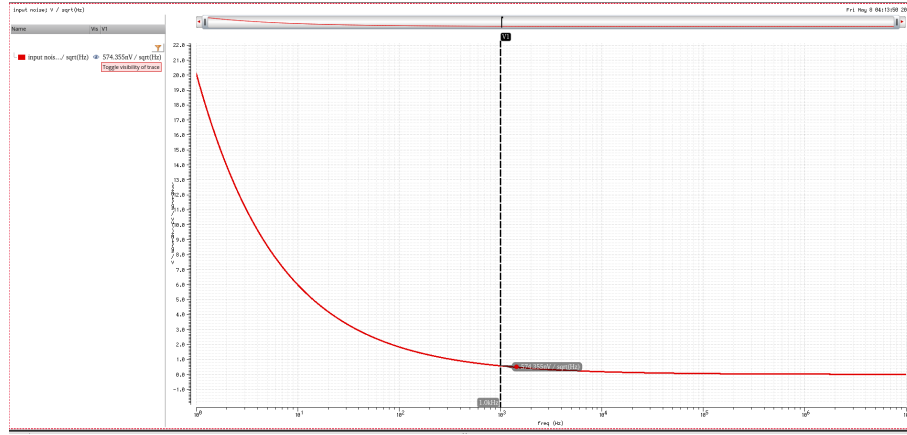
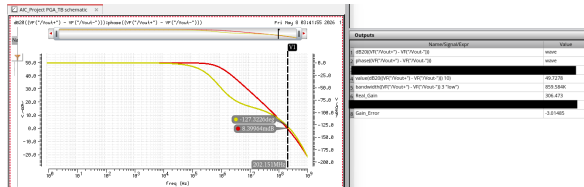


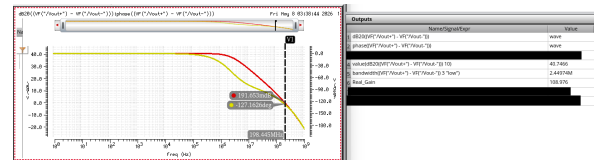
Figure 17: Noise spectrum at the input of the OTA. The input-referred noise at 1 kHz is $574 \text{ nV}/\sqrt{\text{Hz}}$.

The integrated RMS noise value over 1 Hz to 10 MHz is given by the expression $\text{sqrt}(\text{integ}((\text{getData}(\text{"in"} \text{ ?result "noise"})**2) \text{ 1 } 10000000))$. Upon evaluation, the result I got was $81.2\mu\text{V}$.

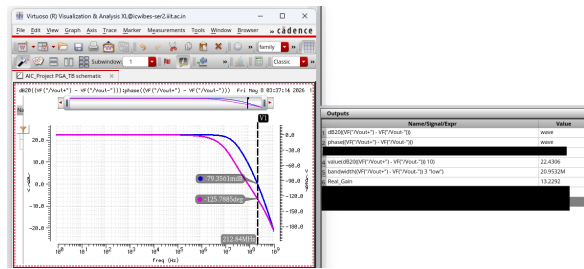
5.7 Programmable Gain Verification



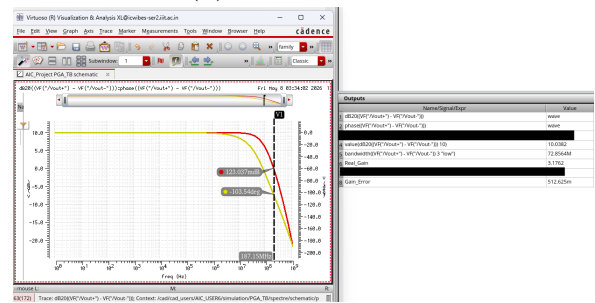
(a) 50 dB setting.



(b) 40 dB setting.



(c) 20 dB setting.



(d) 10 dB setting.

Figure 18: Programmable gain verification for four gain settings.

Table 4: Programmable gain settings and measured performance

Setting	Measured gain (dB)	Measured gain (V/V)	Bandwidth (MHz)	Gain error (%)
50 dB	49.73	306.47	0.860	-3.09
40 dB	40.75	108.98	2.45	—
20 dB	22.43	13.23	20.95	—
10 dB	10.04	3.176	72.86	+0.44

Gain error is computed relative to the ideal linear gains for 10/20/40/50 dB settings.

5.8 PSRR

To find the PSRR, we first find A_{PS} which signifies how much a signal is amplified if the ripple is injected into the power supply. Then, analogous to CMRR, I am using the values at 3 frequencies to get the following results.

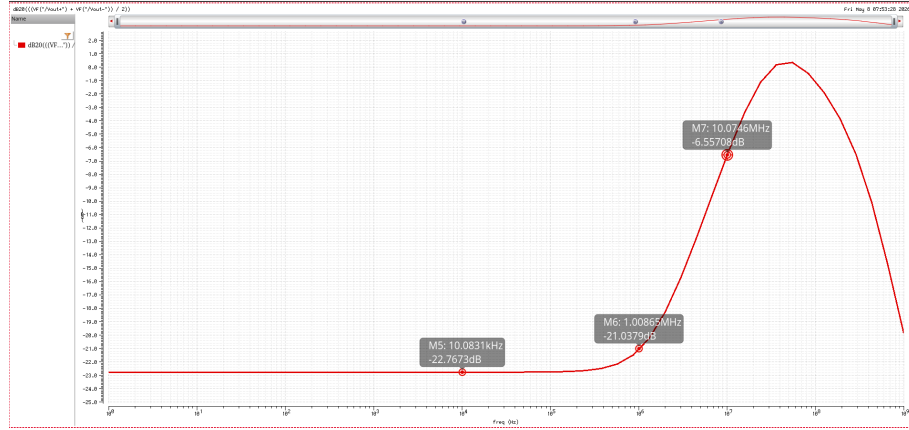


Figure 19: PSRR plotted and marked at 3 different frequencies

Table 5: PSRR extracted from AC analysis

Frequency	A_{dd} (dB)	A_{dm} (dB)	PSRR (dB)
10 kHz	-22.77	69.32	92.09
1 MHz	-21.04	48.23	69.27
10 MHz	-6.56	27.92	34.48

5.9 PVT Analysis

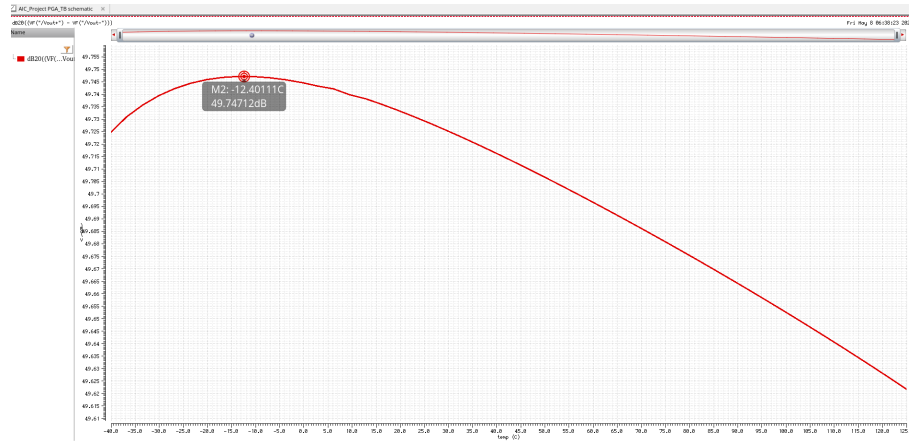


Figure 20: Gain variation at 10 kHz upon sweeping temperature from -40 to 125 degrees. Note that the Y-axis is almost constant upto the first decimal in this plot.

Upon setting different temperature, process and voltage supply corners, this is the plot I observed.

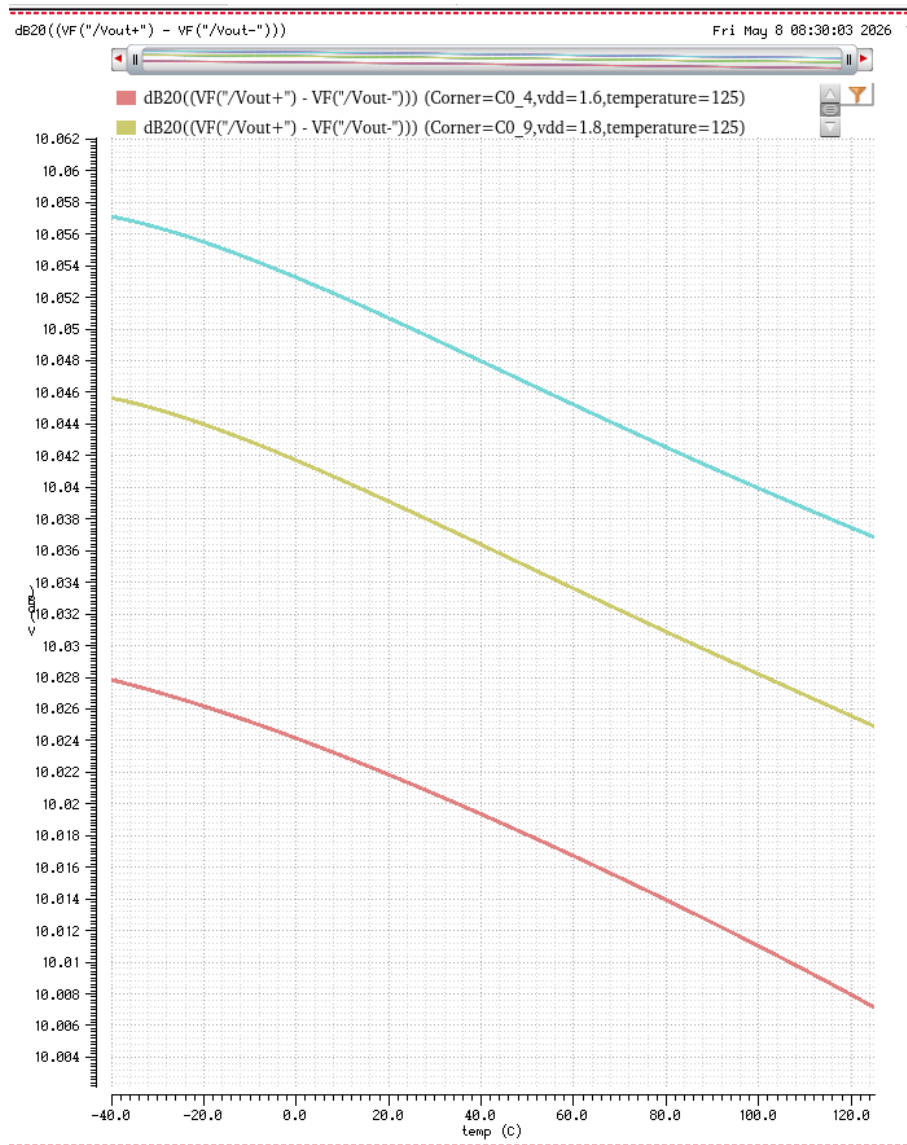


Figure 21: Plot of minimum PGA gain when varied across several temperature (), process (SS, TT, FF) and supply variation corners (1.6, 1.8 and 2.0 V). (The plots for different VDD values overlap here)

6 Summary of the design

Table 6: Design specifications vs. achieved performance

Spec	Target	Achieved (Sim)
Supply voltage	1.8 V	1.8 V
Power (OTA)	≤ 1.0 mW	≈ 1.0 mW
-3dB Bandwidth (OTA)	≥ 75 kHz	89.3 kHz
Gain range (PGA)	10–50 dB	10.04–49.73 dB
Bandwidth at max gain (PGA)	≥ 0.75 MHz	0.86 MHz
Unity-gain bandwidth (OTA)	≥ 237 MHz	245.6 MHz
Open-loop gain (OTA)	70 dB	69.4 dB
Phase margin (OTA)	$\approx 60^\circ$	61.4°
CMFB loop gain	≥ 40 dB	55.56 dB
CMFB phase margin	$\geq 60^\circ$	72°
CMFB bandwidth	5 – 10 MHz	21.75 MHz
Input-referred noise @1 kHz (OTA)	35 nV/ $\sqrt{\text{Hz}}$	574 nV/ $\sqrt{\text{Hz}}$
Slew rate (pos/neg) (PGA)	$\geq 20/20$ V/ μs	$88.3/17$ V/ μs
Gain error (PGA)	$\leq 5\%$	$\leq 3.09\%$
CMRR	≥ 80 KHz	78.74 @ 10 kHz